

1. Data Reporting

1.1. Format as follows

```
1  {
2    "sn": "03000118180807000001",
3    "version": 2,
4    "time": "2025-11-19 10:39:47",
5    "dplist": [
6      {
7        "id": 212,
8        "status": "ReadSucceed",
9        "value": 0,
10       "symbol": "channel-1",
11       "unit": ""
12     },
13     {
14       "id": 215,
15       "status": "ReadFailed",
16       "value": 0,
17       "symbol": "c2",
18       "unit": "mA"
19     }
20   ]
21 }
```

- Possible subcontracted sending scenario
 - The webpage has many point settings
 - The reporting intervals vary across different sites.

1.1. Content Description

- SN: Gateway serial number, which serves as the unique identifier for the gateway
- Version: The version number of the current format. If the format changes later, this number will also update accordingly.
- time: the time when the device sends this data packet
- dplist: Data point information reported, with quantity varying according to settings on the webpage
 - id: A unique ID number for the location, retrieved from the webpage. In Shidian series products, this ID number is unique.
 - status: Current reading status
 - Uncertain: This location data is not updated
 - Readsucceed: The data at this location has been updated

- ReadFailed: Failed to read point data
- value: point reading value
- symbol: A flag set on the webpage. If not set, it is an empty value ("").
- Unit: The unit set in the webpage. If not set, it defaults to empty ("").

2. Data distribution

2.1 Send

```

1  {
2      "sn": "03000118180807000001",
3      "version": 2,
4      "seq": 1022,
5      "dplist": [
6          {
7              "id": 214,
8              "value": 1
9          },
10         {
11             "id": 215,
12             "value": 0
13         }
14     ]
15 }
```

- A single package supports up to 16 data points.

2.1.1 Content Description

- sn: Unique device identifier that must match the device
- Version: Format version number for supporting multiple version formats
- seq: Data frame sequence number, included in the response frame to facilitate message correspondence
- dplist: Content for points to be set
 - id: Unique identifier for the location, retrieved from the webpage
 - value: the value to set
 - Note: The device is not calibrated for this range. Setting an invalid value may cause errors.
 - For example, the valid range for point location settings is 0 to 100, but users may enter values outside this range.

2.2 Reply

2.2.1 Normal Reply Format

```
1  {
2    "sn": "03000118180807000001",
3    "version": 2,
4    "seq": 1022,
5    "time": "2025-11-19 11:22:01",
6    "res": "ok"
7  }
```

2.2.2 Abnormal Response Format

```
1  {
2    "sn": "03000118180807000001",
3    "version": 2,
4    "seq": 0,
5    "time": "2025-11-19 11:23:11",
6    "res": "parse error"
7  }
```

```
1  {
2    "sn": "03000118180807000001",
3    "version": 2,
4    "seq": 1022,
5    "time": "2025-11-19 13:07:22",
6    "res": "warning",
7    "fail list": [
8      218,
9      219
10   ]
11 }
```

2.2.3 Content Description

- seq: The sequence number from the sending frame. If the sending frame is abnormal or not set, it defaults to 0.
- res: Current settings result

Results	explain
ok	The device is correct and has parsed the data packet. The location is valid.
warning	Some points were rejected, possibly due to ID errors, failure to bind to the northbound direction, or missing values.
parse error	Packet parsing failed. The issue is likely due to incorrect JSON format.
sn error	The serial number is incorrect and does not match the device.
version error	The version is not supported
point parse error	Failed to retrieve location information
point num error	The number of points exceeds 16, which is beyond the single processing capacity. The entire data package will not be processed.
other error	Other unknown causes

● fail list:

- The format is correct, but the device rejected the point list.